Database structure

The database will consist of related tables rather than one large spreadsheet. Each record will have a unique primary key, and foreign keys will connect related records.

Here is the initial design.

A. Employee and asset management

Purpose: Track staff, inventory and asset assignments.

`employees`

- `employee_id` — Primary key
- `employee_number` — Unique employee number
- `full_name`
- `department`
- `designation`
- `contact_number`
- `email`
- `is_active`

`assets`

- `asset_id` — Primary key
- `asset_code` — Manually assigned, unique
- `asset_name`
- `category_id` — Foreign key
- `serial_number`
- `purchase_order_id` — Foreign key, optional
- `purchase_date`
- `purchase_cost`
- `asset_status`
- `current_location`

`asset_categories`

- `category_id` — Primary key
- `category_name`
- `description`

B. Procurement and contracts

Purpose: Link purchases, vendors, contracts and invoices.

`vendors`

- vendor_id — Primary key
- vendor_name
- registration_number
- contact_person
- phone
- email
- address

purchase_orders

- purchase_order_id — Primary key
- po_number — Unique
- vendor_id — Foreign key
- contract_id — Optional foreign key
- po_date
- total_amount
- currency
- status
- document_path

contracts

- contract_id — Primary key
- contract_number — Unique
- vendor_id — Foreign key
- start_date
- expiry_date
- contract_value
- renewal_status
- document_path

invoices

- invoice_id — Primary key
- invoice_number
- purchase_order_id — Foreign key
- invoice_date
- invoice_amount
- payment_status
- payment_date

C. Issuance and returns

Purpose: Maintain a complete history of asset movements.

asset_transactions

- transaction_id — Primary key

- asset_id — Foreign key
- employee_id — Foreign key
- transaction_type — Issue or return
- transaction_date
- condition_notes
- reference_number
- processed_by
- remarks

Important: Asset issuance and return forms will be generated from these records. We will also design document-header and document-line tables where a single form covers multiple assets.

D. Maintenance and software

maintenance_records

- maintenance_id — Primary key
- asset_id — Foreign key, optional
- contract_id — Foreign key, optional
- vendor_id — Foreign key, optional
- maintenance_type
- request_date
- completion_date
- cost
- status
- remarks

software_licenses

- license_id — Primary key
- software_name
- license_key_reference — Secure reference, not a publicly displayed key
- vendor_id — Foreign key, optional
- purchase_order_id — Foreign key, optional
- start_date
- expiry_date
- license_quantity
- renewal_cost
- status

E. Annual budget

financial_years

- financial_year_id — Primary key
- year_name
- start_date

- `end_date`
- `status`

`budget_allocations`

- `budget_id` — Primary key
- `financial_year_id` — Foreign key
- `budget_category`
- `allocated_amount`
- `currency`

`budget_transactions`

- `budget_transaction_id` — Primary key
- `budget_id` — Foreign key
- `purchase_order_id` — Optional foreign key
- `invoice_id` — Optional foreign key
- `transaction_type`
- `amount`
- `transaction_date`
- `remarks`

Key design decisions:

- PO and contract: Linked but independently recorded; a PO may refer to a contract, while a contract may cover multiple POs.
- PO line items: A PO can contain many different items or services.
- Invoices: A PO may have multiple invoices, including partial invoices.
- Assets: Each physical asset gets a manually assigned unique asset code.
- Issuance and return: Movement history is retained rather than overwritten.
- Budget: Allocations and financial transactions are recorded separately to prevent double counting.
- Documents: Existing templates are treated as controlled document templates; the database stores their data and references to generated files.